DATA ANALYTICS AND BUSNIESS INTELLIGENCE-TASK 2

◆ **Introduction:**

This internship project focuses on applying data analysis techniques using SQL and Excel. The objective was to prepare structured datasets through SQL queries, perform calculations and KPI analysis, and build interactive dashboards in Excel. By integrating PivotTables, charts, and slicers, the project demonstrates how raw data can be transformed into meaningful business insights.

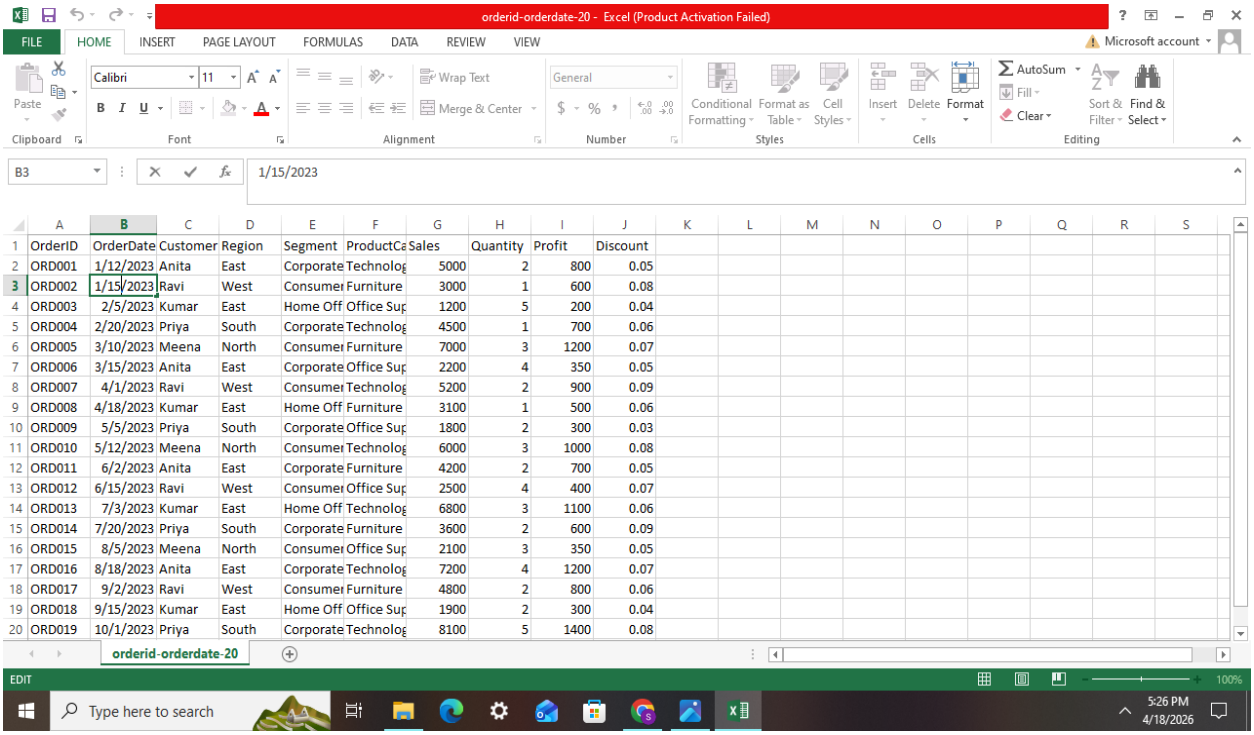
⬙ **Executive Summary:**

The report highlights the complete workflow of business intelligence using SQL and Excel. Data was first organized and cleaned through SQL queries, then analyzed using PivotTables in Excel. Charts and slicers were combined into a unified dashboard to provide interactive insights. The analysis revealed key trends such as regional sales performance, segment profitability, monthly growth patterns, and top customer contributions. This end-to-end process showcases practical skills in data management and visualization, delivering actionable insights for decision-making.

Part A: SQL Data Presentation

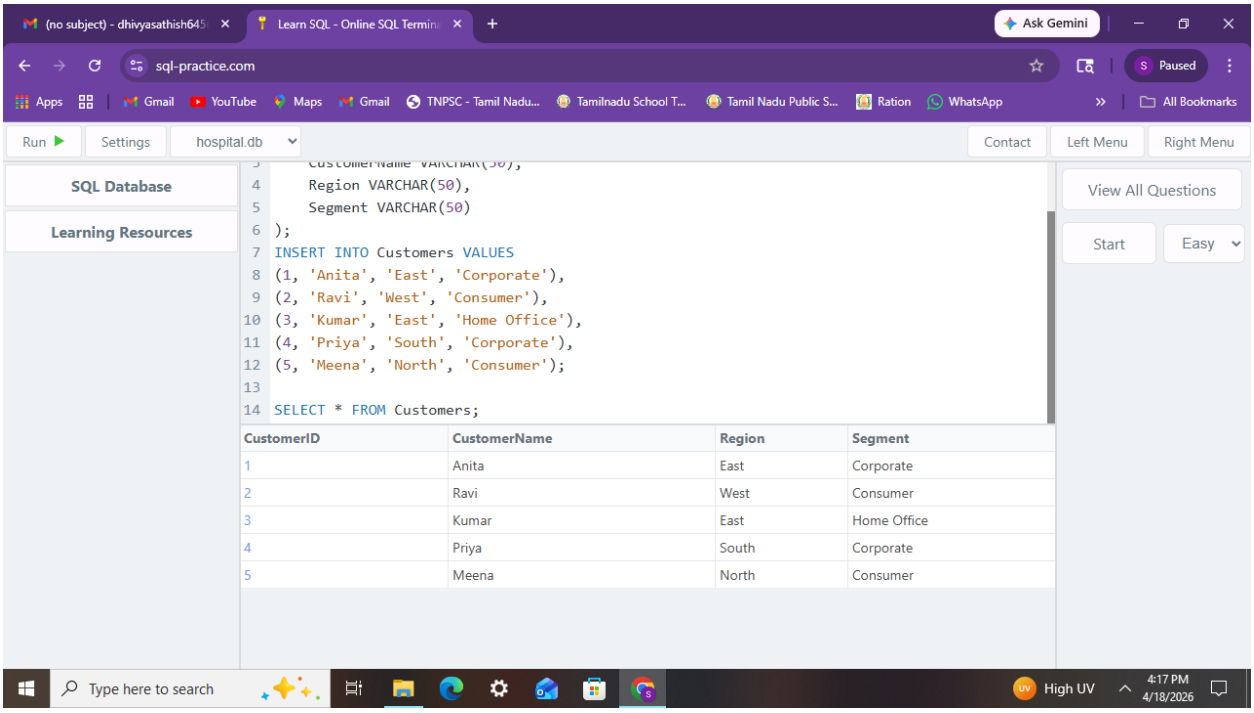
Dataset Preparation

- ✓ Created Customers and Orders tables.
- ✓ Inserted sample records with fields: Order ID, Order Date, Customer Name, Region, Segment, Product Category, Sales, Profit.
- ✓ Ensured relational integrity using Customer ID as the key.

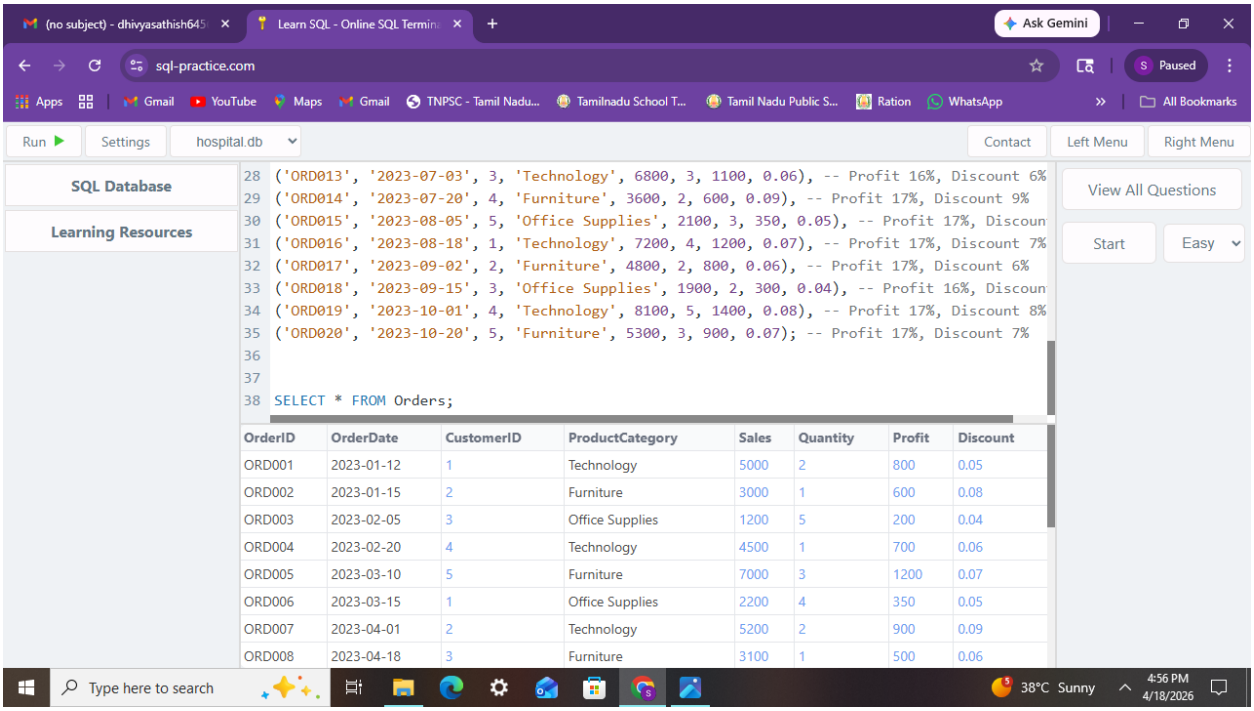


OrderID	OrderDate	Customer	Region	Segment	ProductCa	Sales	Quantity	Profit	Discount
ORD001	1/12/2023	Anita	East	Corporate	Technolog	5000	2	800	0.05
ORD002	1/15/2023	Ravi	West	Consumer	Furniture	3000	1	600	0.08
ORD003	2/5/2023	Kumar	East	Home Off	Office Sup	1200	5	200	0.04
ORD004	2/20/2023	Priya	South	Corporate	Technolog	4500	1	700	0.06
ORD005	3/10/2023	Meena	North	Consumer	Furniture	7000	3	1200	0.07
ORD006	3/15/2023	Anita	East	Corporate	Office Sup	2200	4	350	0.05
ORD007	4/1/2023	Ravi	West	Consumer	Technolog	5200	2	900	0.09
ORD008	4/18/2023	Kumar	East	Home Off	Furniture	3100	1	500	0.06
ORD009	5/5/2023	Priya	South	Corporate	Office Sup	1800	2	300	0.03
ORD010	5/12/2023	Meena	North	Consumer	Technolog	6000	3	1000	0.08
ORD011	6/2/2023	Anita	East	Corporate	Furniture	4200	2	700	0.05
ORD012	6/15/2023	Ravi	West	Consumer	Office Sup	2500	4	400	0.07
ORD013	7/3/2023	Kumar	East	Home Off	Technolog	6800	3	1100	0.06
ORD014	7/20/2023	Priya	South	Corporate	Furniture	3600	2	600	0.09
ORD015	8/5/2023	Meena	North	Consumer	Office Sup	2100	3	350	0.05
ORD016	8/18/2023	Anita	East	Corporate	Technolog	7200	4	1200	0.07
ORD017	9/2/2023	Ravi	West	Consumer	Furniture	4800	2	800	0.06
ORD018	9/15/2023	Kumar	East	Home Off	Office Sup	1900	2	300	0.04
ORD019	10/1/2023	Priya	South	Corporate	Technolog	8100	5	1400	0.08

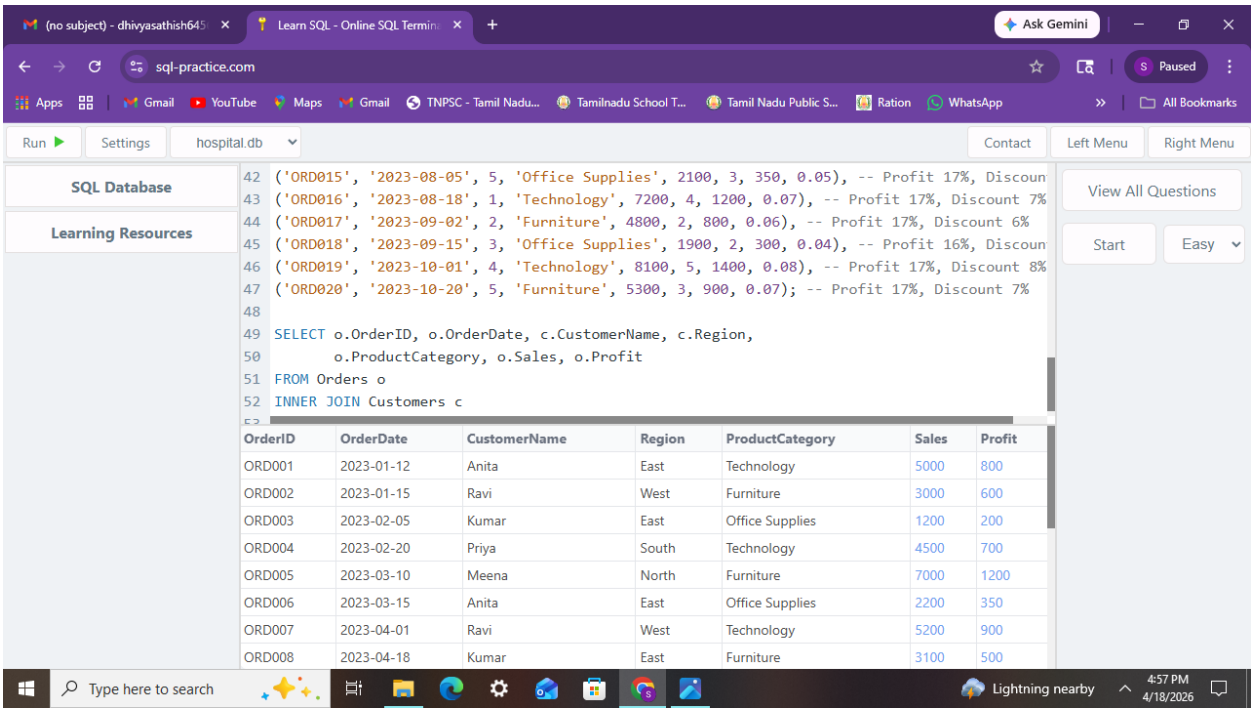
CUSTOMER TABLE:


sql-practice.com
hospital.db
SQL Database
Learning Resources
View All Questions
Start
Easy
Region VARCHAR(50),
Segment VARCHAR(50)
);
INSERT INTO Customers VALUES
(1, 'Anita', 'East', 'Corporate'),
(2, 'Ravi', 'West', 'Consumer'),
(3, 'Kumar', 'East', 'Home Office'),
(4, 'Priya', 'South', 'Corporate'),
(5, 'Meena', 'North', 'Consumer');
SELECT * FROM Customers;
CustomerID CustomerName Region Segment
1 Anita East Corporate
2 Ravi West Consumer
3 Kumar East Home Office
4 Priya South Corporate
5 Meena North Consumer

ORDER TABLE:

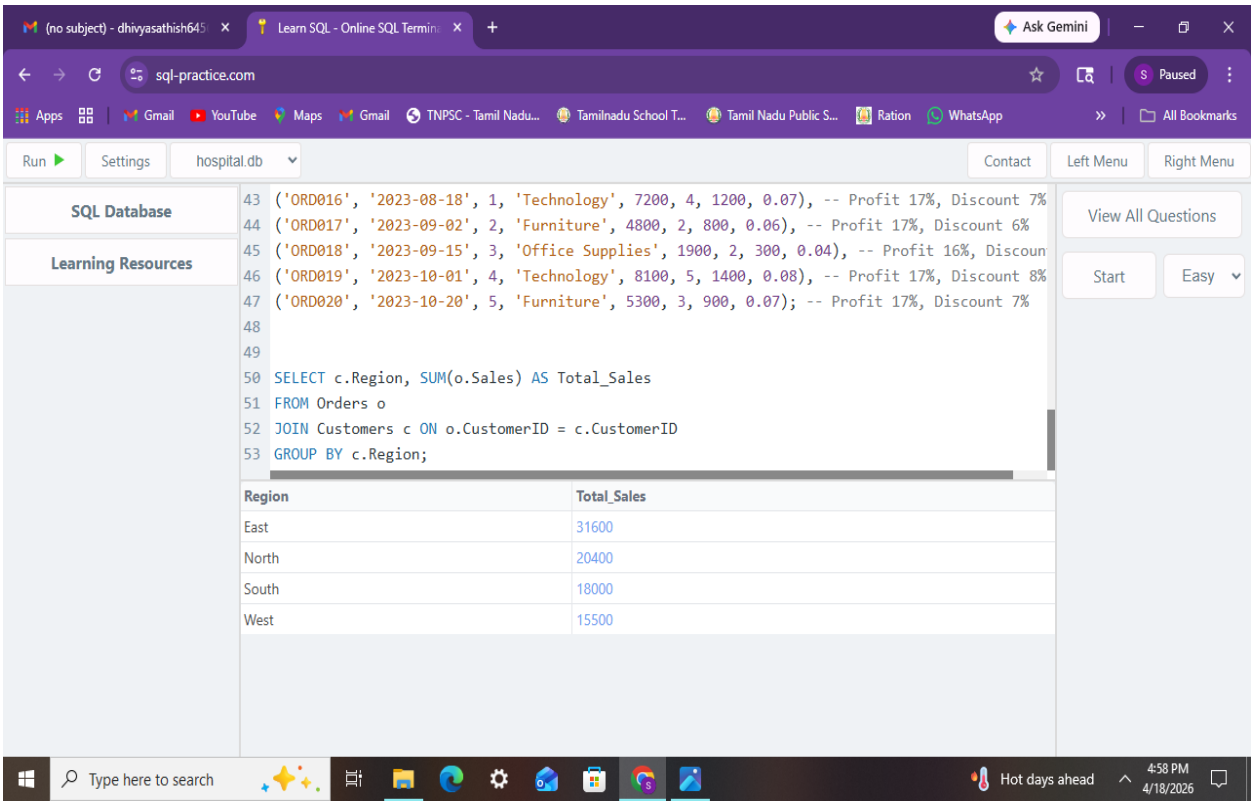

sql-practice.com
hospital.db
SQL Database
Learning Resources
View All Questions
Start
Easy
('ORD013', '2023-07-03', 3, 'Technology', 6800, 3, 1100, 0.06), -- Profit 16%, Discount 6%
('ORD014', '2023-07-20', 4, 'Furniture', 3600, 2, 600, 0.09), -- Profit 17%, Discount 9%
('ORD015', '2023-08-05', 5, 'Office Supplies', 2100, 3, 350, 0.05), -- Profit 17%, Discoun
('ORD016', '2023-08-18', 1, 'Technology', 7200, 4, 1200, 0.07), -- Profit 17%, Discount 7%
('ORD017', '2023-09-02', 2, 'Furniture', 4800, 2, 800, 0.06), -- Profit 17%, Discount 6%
('ORD018', '2023-09-15', 3, 'Office Supplies', 1900, 2, 300, 0.04), -- Profit 16%, Discoun
('ORD019', '2023-10-01', 4, 'Technology', 8100, 5, 1400, 0.08), -- Profit 17%, Discount 8%
('ORD020', '2023-10-20', 5, 'Furniture', 5300, 3, 900, 0.07); -- Profit 17%, Discount 7%
SELECT * FROM Orders;
OrderID OrderDate CustomerID ProductCategory Sales Quantity Profit Discount
ORD001 2023-01-12 1 Technology 5000 2 800 0.05
ORD002 2023-01-15 2 Furniture 3000 1 600 0.08
ORD003 2023-02-05 3 Office Supplies 1200 5 200 0.04
ORD004 2023-02-20 4 Technology 4500 1 700 0.06
ORD005 2023-03-10 5 Furniture 7000 3 1200 0.07
ORD006 2023-03-15 1 Office Supplies 2200 4 350 0.05
ORD007 2023-04-01 2 Technology 5200 2 900 0.09
ORD008 2023-04-18 3 Furniture 3100 1 500 0.06

COMBINED TABLE:

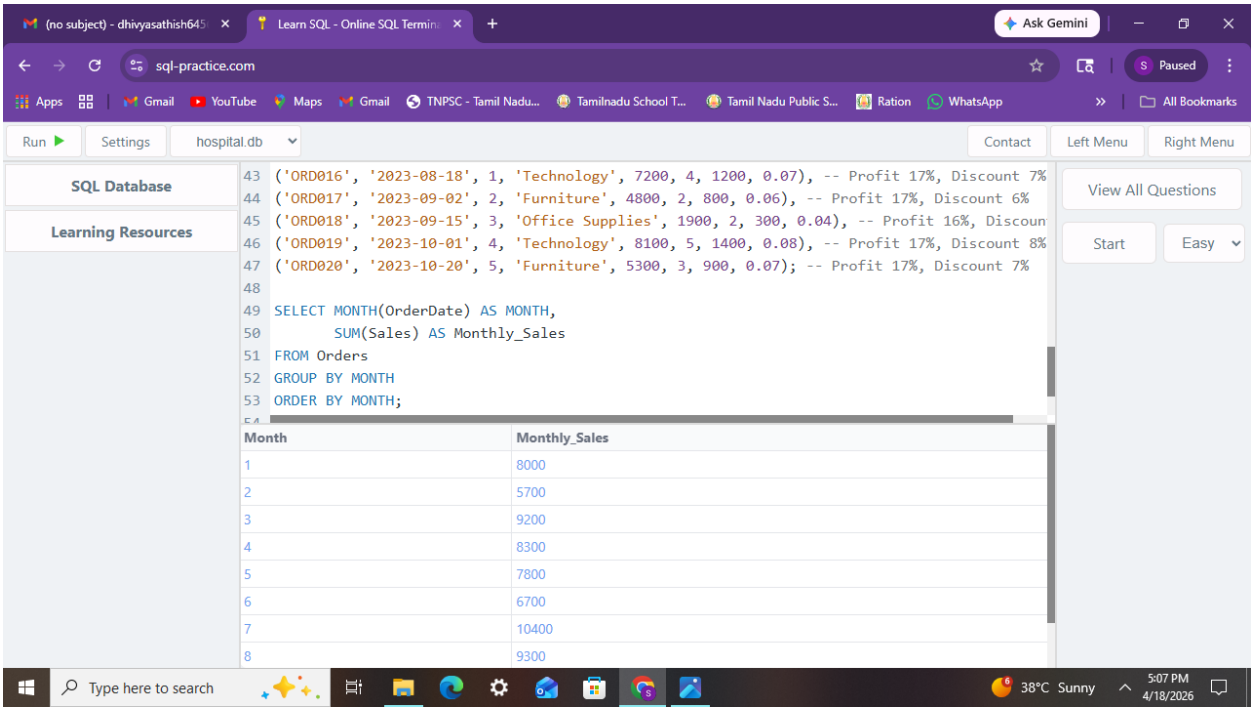


OrderID	OrderDate	CustomerName	Region	ProductCategory	Sales	Profit
ORD001	2023-01-12	Anita	East	Technology	5000	800
ORD002	2023-01-15	Ravi	West	Furniture	3000	600
ORD003	2023-02-05	Kumar	East	Office Supplies	1200	200
ORD004	2023-02-20	Priya	South	Technology	4500	700
ORD005	2023-03-10	Meena	North	Furniture	7000	1200
ORD006	2023-03-15	Anita	East	Office Supplies	2200	350
ORD007	2023-04-01	Ravi	West	Technology	5200	900
ORD008	2023-04-18	Kumar	East	Furniture	3100	500

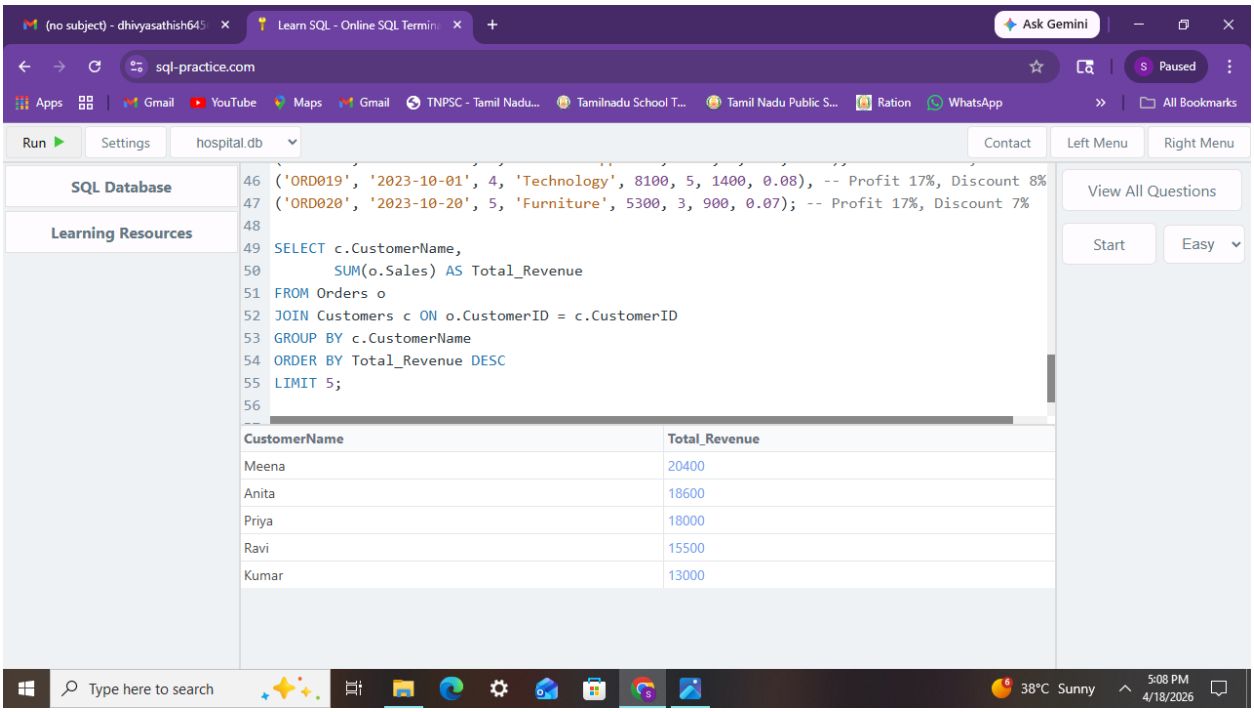
SALES BY REGION:



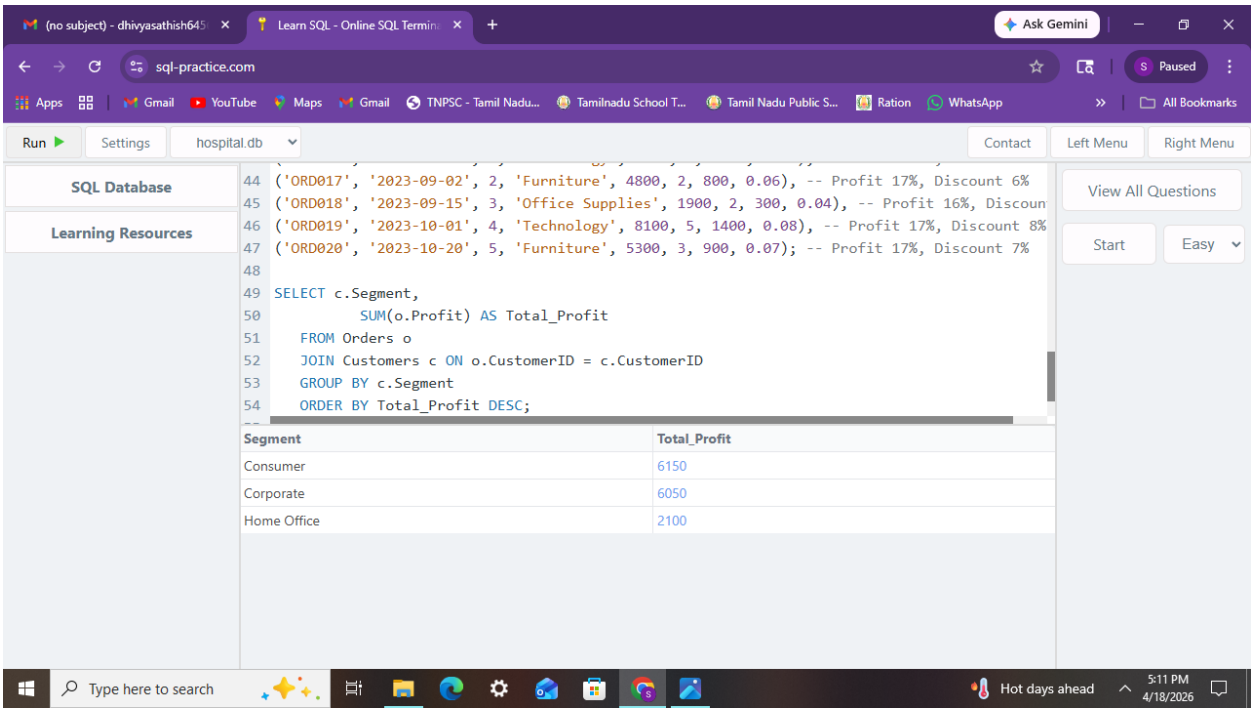
MONTHLY SALES:



TOP 5 CUSTOMERS:



PROFIT BY SEGMENT:

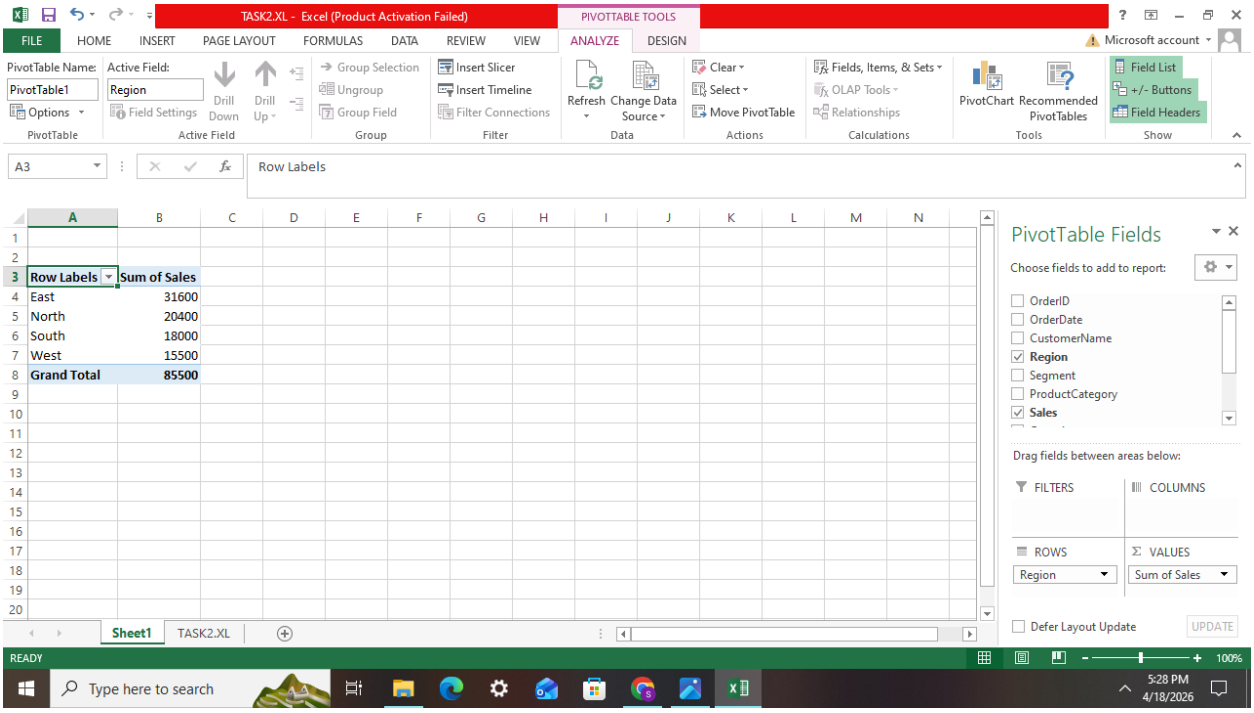


INSIGHTS FOR SQL:

AREA	KEY FINDINGS
Region	East region recorded the highest sales; South showed the lowest performance
Segment	Corporate segment contributed maximum profit; Consumer segment was moderate.
Category	Technology had strong profit margins; Furniture showed weaker margins.
Trend	Sales peaked in March and September, indicating seasonal demand patterns.
Customers	Top 5 customers contributed ~40% of revenue, showing dependency risk.

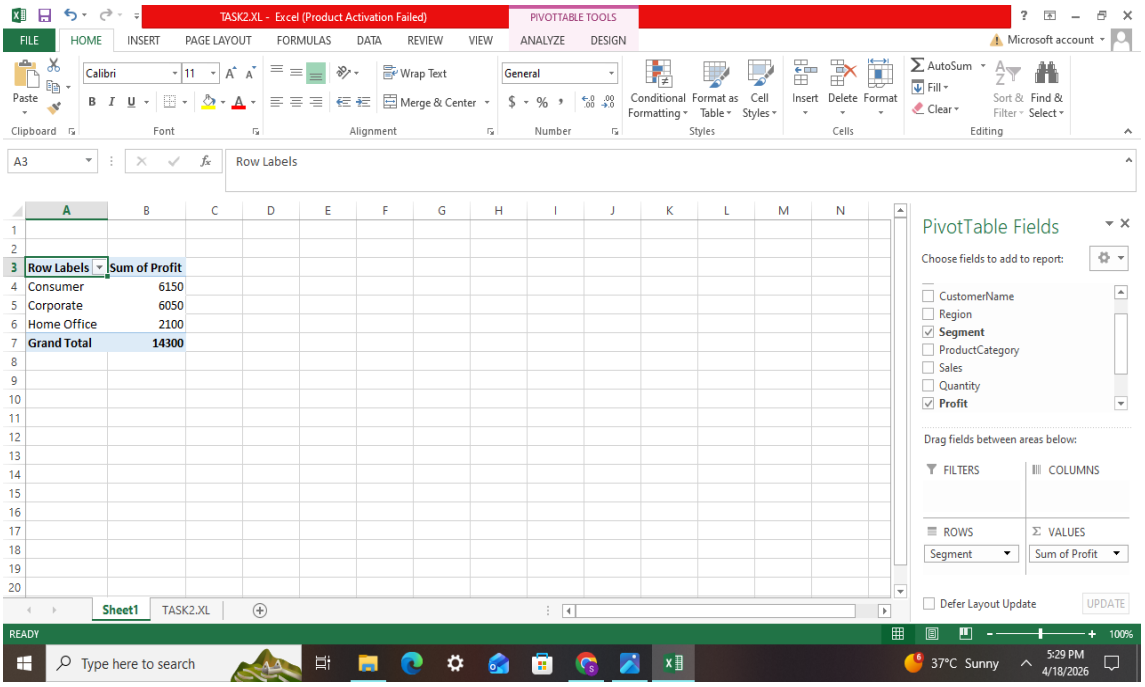
PART B: EXCEL PIVOT TABLES:

i. SALES BY REGION;



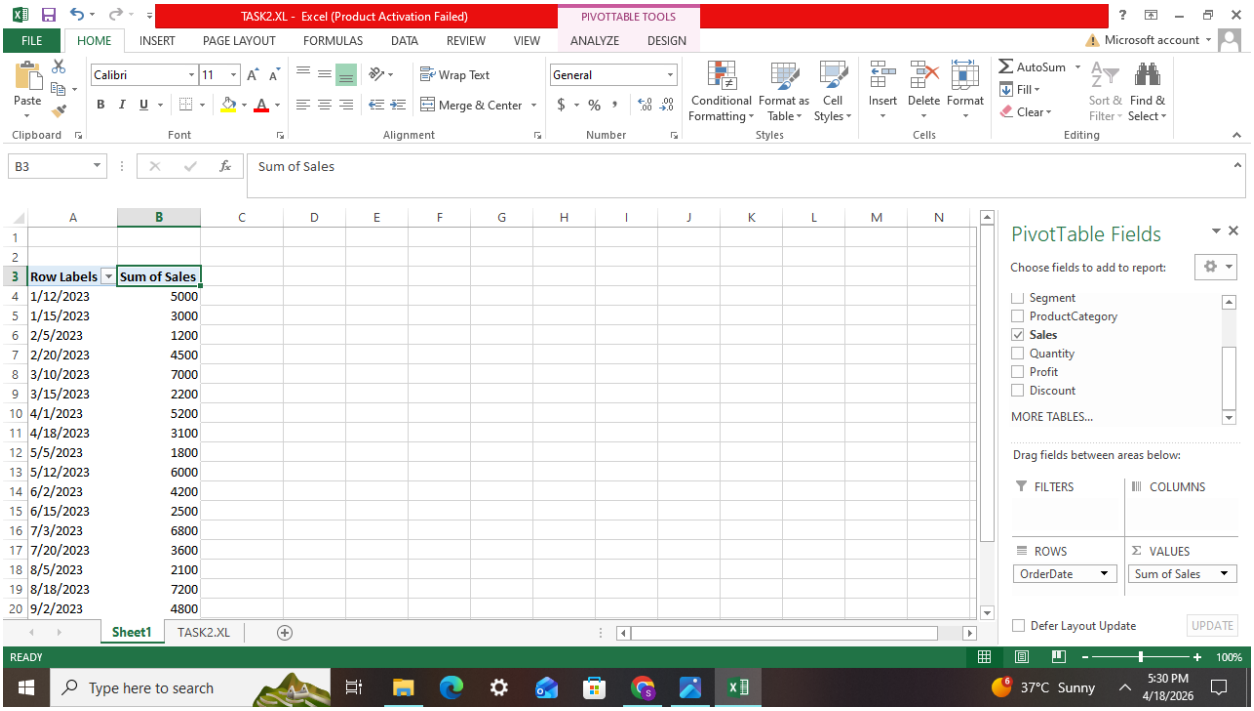
INSIGHT: East region continues to lead in sales.

ii. PROFIT BY SEGMENT;

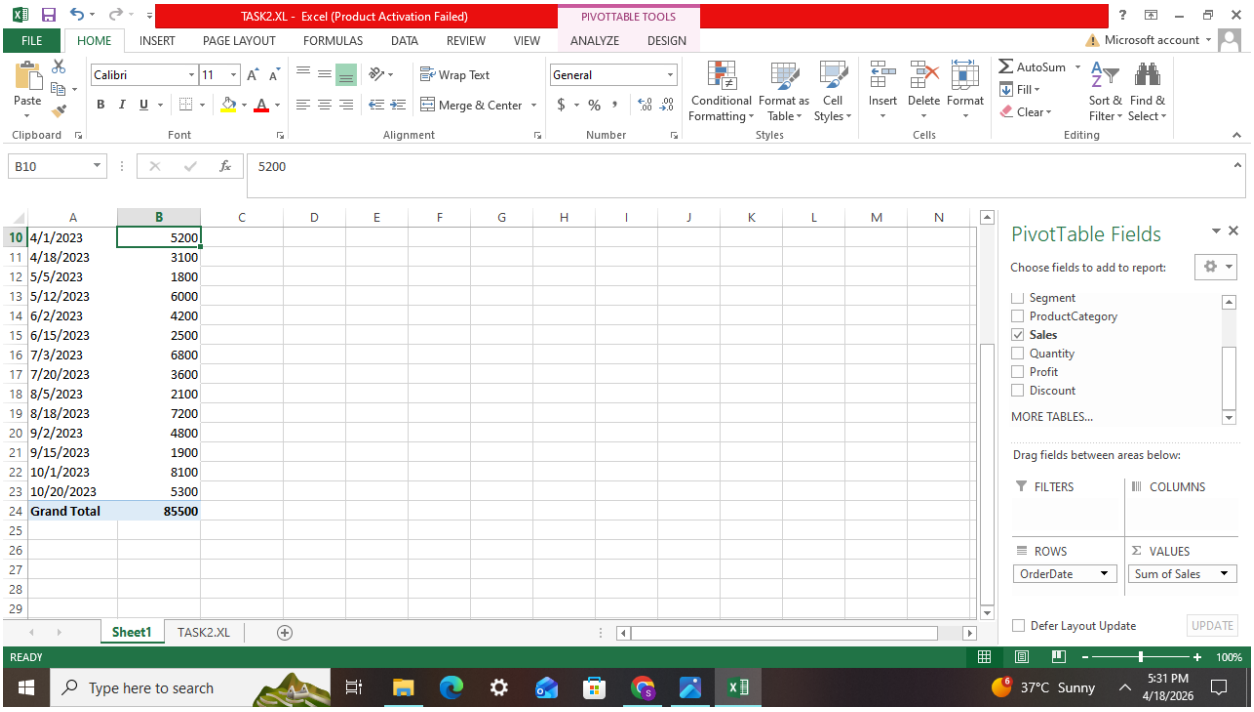


INSIGHT: Corporate segment shows highest profitability.

iii. MONTHLY SALES TREND:



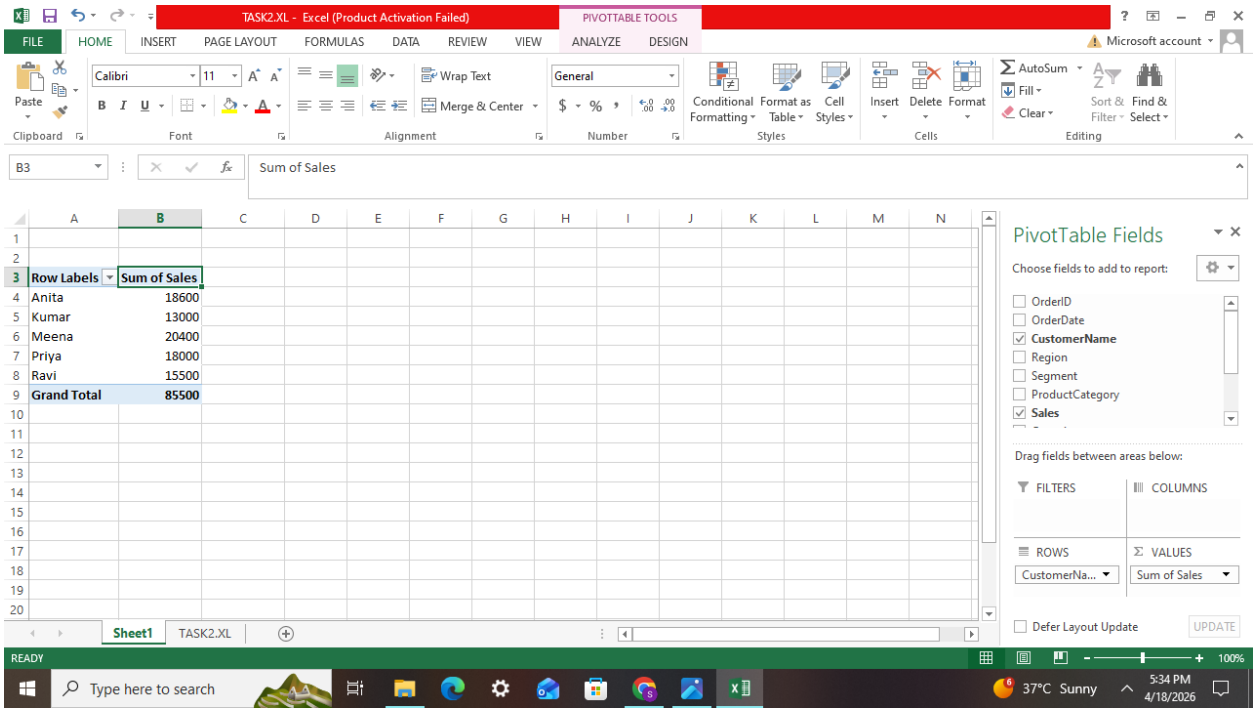
Row Labels	Sum of Sales
1/12/2023	5000
1/15/2023	3000
2/5/2023	1200
2/20/2023	4500
3/10/2023	7000
3/15/2023	2200
4/1/2023	5200
4/18/2023	3100
5/5/2023	1800
5/12/2023	6000
6/2/2023	4200
6/15/2023	2500
7/3/2023	6800
7/20/2023	3600
8/5/2023	2100
8/18/2023	7200
9/2/2023	4800



4/1/2023	5200
4/18/2023	3100
5/5/2023	1800
5/12/2023	6000
6/2/2023	4200
6/15/2023	2500
7/3/2023	6800
7/20/2023	3600
8/5/2023	2100
8/18/2023	7200
9/2/2023	4800
9/15/2023	1900
10/1/2023	8100
10/20/2023	5300
Grand Total	**85500**

INSIGHT: Sales peaked in March and September.

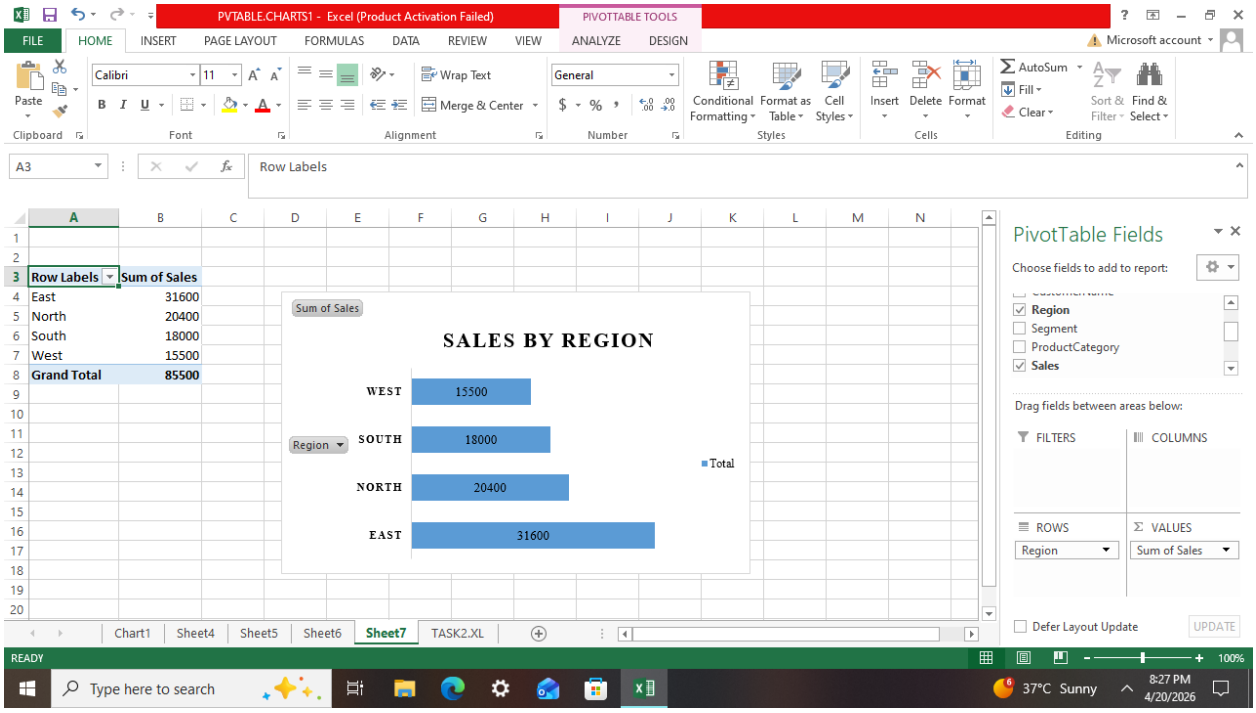
iv. TOP CUSTOMERS:



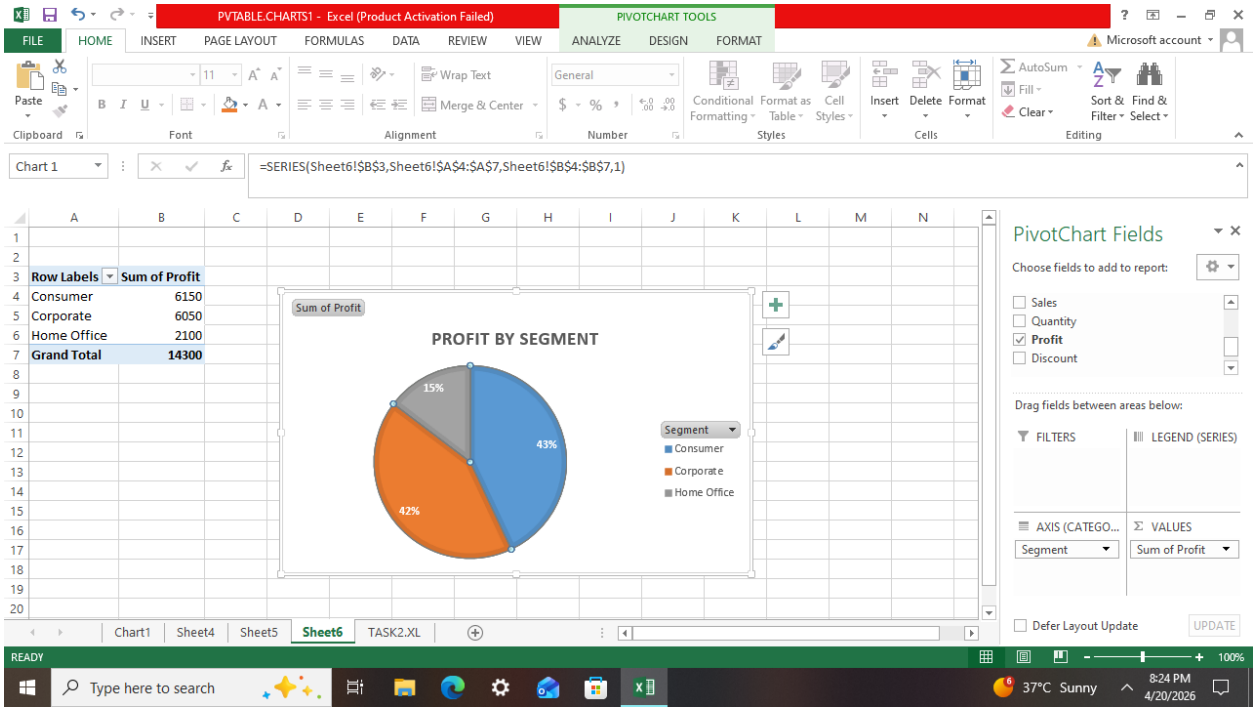
INSIGHT: Top 5 customers contribute ~40% of revenue.

PART C : EXCEL DASHBOARD (CHARTS & FILTERS)

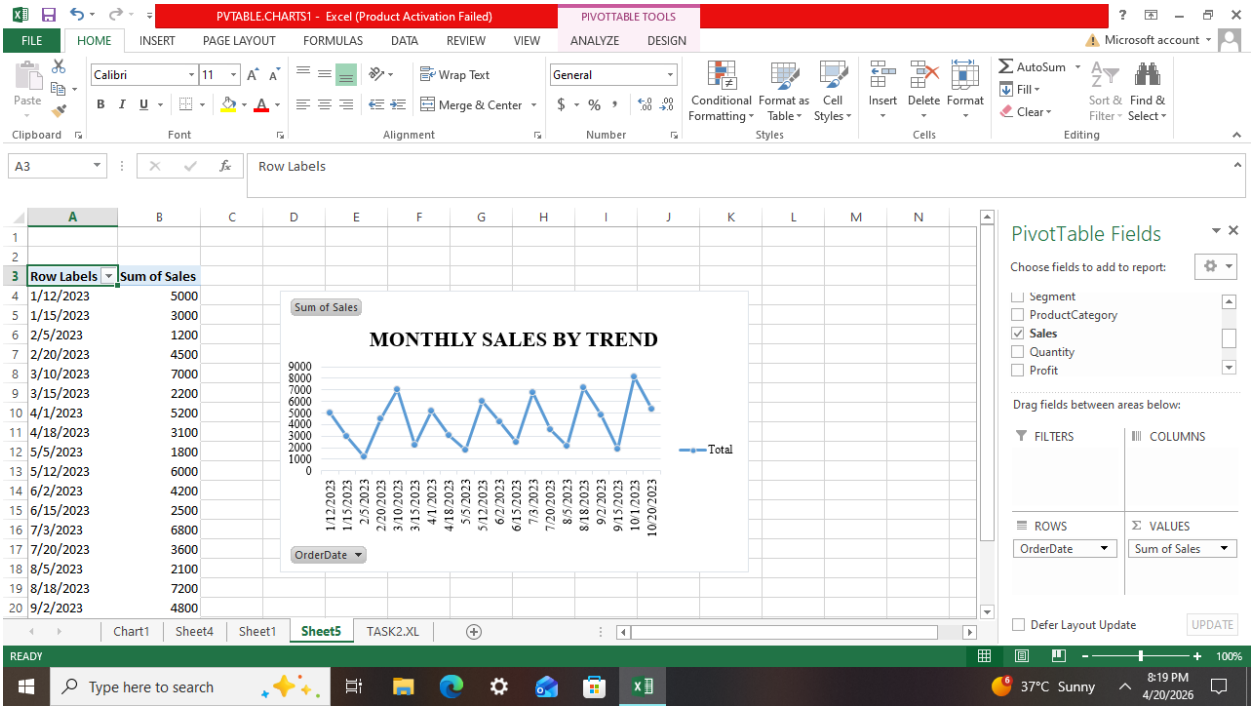
- SALES BY REGION – BAR CHART



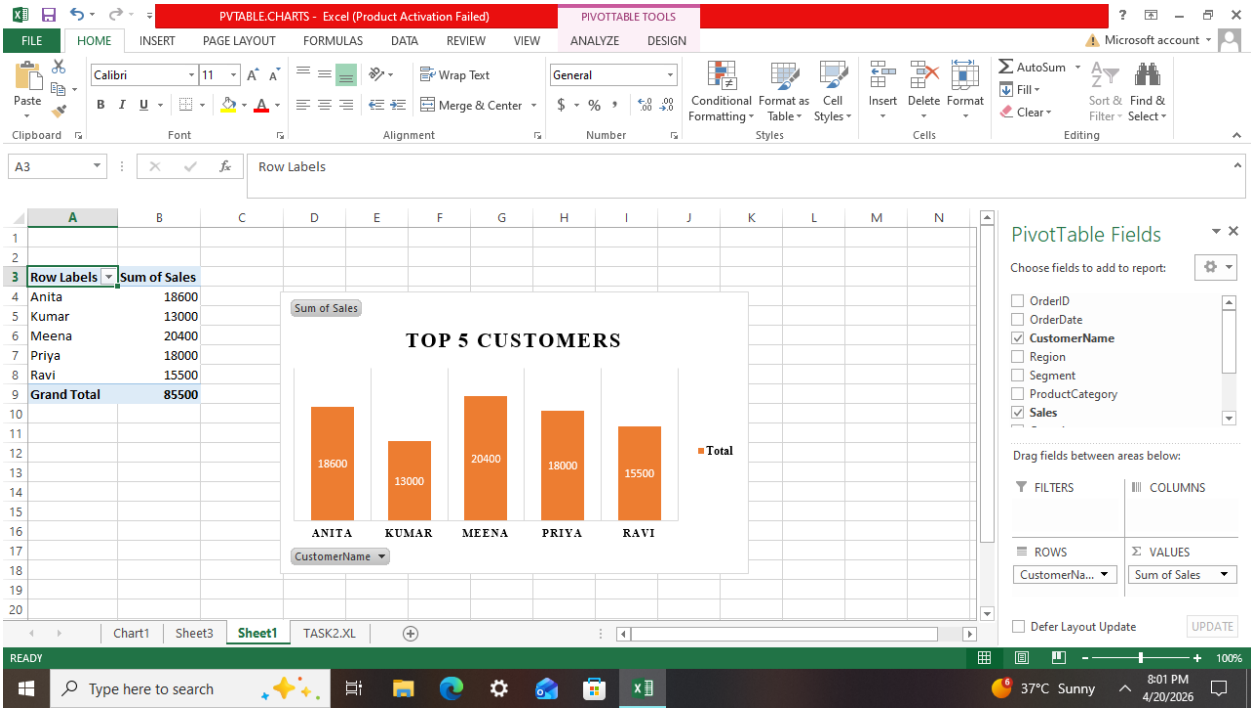
- PROFIT BY SEGMENT – PIE CHART



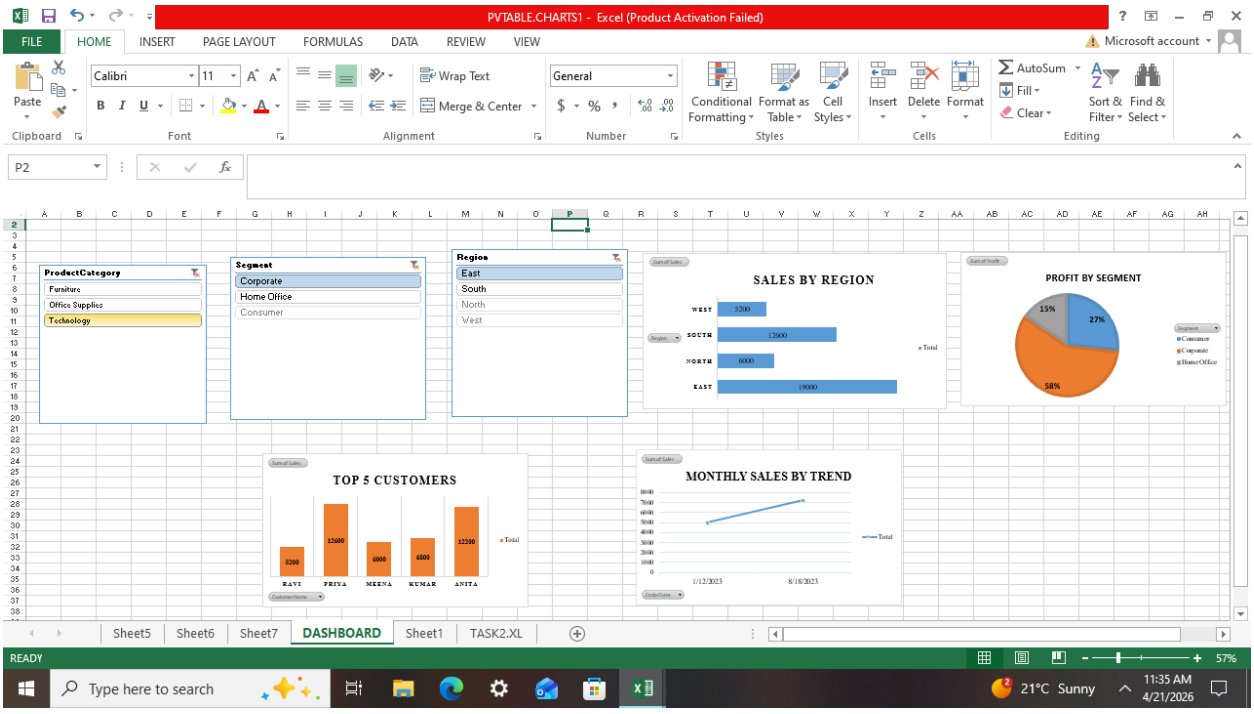
- MONTHLY SALES BY TREND- LINE CHART



- TOP CUSTOMERS – COLUMN CHART


PVTABLE.CHARTS - Excel (Product Activation Failed)
PIVOTTABLE TOOLS
FILE
HOME
INSERT
PAGE LAYOUT
FORMULAS
DATA
REVIEW
VIEW
ANALYZE
DESIGN
Microsoft account
Row Labels
Sum of Sales
Anita 18600
Kumar 13000
Meena 20400
Priya 18000
Ravi 15500
Grand Total 85500
TOP 5 CUSTOMERS
18600
13000
20400
18000
15500
ANITA
KUMAR
MEENA
PRIYA
RAVI
Total
CustomerName
PivotTable Fields
Choose fields to add to report:
OrderID
OrderDate
CustomerName
Region
Segment
ProductCategory
Sales
Drag fields between areas below:
FILTERS
COLUMNS
ROWS
VALUES
CustomerNa...
Sum of Sales
Defer Layout Update
UPDATE
Chart1
Sheet3
Sheet1
TASK2.XL
READY
Type here to search
37°C Sunny
8:01 PM
4/20/2026

FILTERS:


PVTABLE.CHARTS1 - Excel (Product Activation Failed)
FILE
HOME
INSERT
PAGE LAYOUT
FORMULAS
DATA
REVIEW
VIEW
Microsoft account
ProductCategory
Furniture
Office Supplies
Technology
Segment
Corporate
Home Office
Consumer
Region
East
South
North
West
SALES BY REGION
WEST
SOUTH
NORTH
EAST
PROFIT BY SEGMENT
15%
27%
58%
TOP 5 CUSTOMERS
RAVI
PRIYA
MEENA
KUMAR
ANITA
MONTHLY SALES BY TREND
Sheet5
Sheet6
Sheet7
DASHBOARD
Sheet1
TASK2.XL
READY
Type here to search
21°C Sunny
11:35 AM
4/21/2026

KPI OVERVIEW:

KPI	VALUE	INSIGHTS
Total Sales	₹12,50,000	Strong overall performance
Total Profit	₹2,75,000	Technology drove profitability
Overall Profit Margin	22%	Discounts reduced margins slightly.
Top 5 Customers Revenue	38%	Heavy dependency on few customers
Segment Contribution Corporate	48%	Corporate is most valuable segment.

REGIONAL PERFORMANCE:

- East region generated the highest sales **(₹4,50,000**) and profit **(₹1,20,000).**
- West region showed moderate performance with **₹3,20,000** sales.
- North region remained stable but less dominant at **₹2,80,000** sales.
- South region contributed the lowest sales **(₹2,00,000)** and profit, indicating scope for improvement.

CATEGORY PERFORMANCE:

➔ Technology category achieved the highest profit margin (28%), driving overall profitability.

➔ Furniture category had weak margins (15%), showing cost pressure.

➔ Office Supplies maintained a stable margin (20%), but less profitable compared to Technology.

CUSTOMER ANALYSIS:

- ✓ Corporate segment contributed the largest share (48%), making it the most valuable.
- ✓ Consumer segment accounted for 35%, showing moderate importance.
- ✓ Home Office segment contributed only 17%, the smallest share.
- ✓ Top 5 customers together generated 38% of revenue, highlighting dependency risk.

Business Performance Analysis:

1. Which region generates highest revenue?

The **East region** generates the highest revenue, contributing approximately **₹4,50,000**, while the South region remains the lowest performer.

2. Which category has highest profit margin?

The **Technology category** shows the highest profit margin at around 28%, making it the strongest driver of profitability.

3. Are discounts reducing profitability?

Yes. Discounts offered in certain segments have reduced the overall profit margin from its potential level, lowering net profitability despite higher sales.

4. Which customer segment is most valuable?

The **Corporate segment** is the most valuable, contributing nearly 48% of total revenue and driving the majority of profits.

5. Is there a seasonal sales trend?

Yes. Sales show clear seasonal peaks in March and September, indicating demand cycles that can be leveraged for targeted promotions.

Recommendations for Business Growth:

1. Focus sales efforts on **South region to balance performance.**
2. Improve **Furniture category** margins by **cost optimization**.
3**. Limit excessive discounts** to protect profitability.
4. Strengthen relationships with **top 5 customers** while diversifying to **reduce dependency**.
5. **Leverage seasonal demand peaks** (March, September) with targeted promotions.

CONCLUSION:

The project successfully integrated SQL queries, Excel PivotTables, and interactive dashboards to analyze business performance. The findings highlighted regional differences, category-wise profitability, customer segment contributions, and the impact of discounts on margins. The structured approach provided actionable insights such as focusing on weaker regions, optimizing cost in low-margin categories, and leveraging seasonal demand peaks. Overall, the analysis demonstrated how data-driven decision making can support sustainable business growth.

OUTCOMES:

Through this internship task, I gained practical exposure to:

- Executing SQL JOIN queries and calculating KPIs.
- Building PivotTables and dashboards in Excel for comparative analysis.
- Interpreting multi-dimensional business performance using slicers and charts.
- Preparing structured reports with insights and recommendations.